

Modern academy

Project of Discrete Mathematics

Class:C3,C1

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THE Equation is

$$\neg[p \rightarrow (\neg q \wedge \neg r)] \equiv p \wedge [(q \rightarrow r) \rightarrow r]$$

Proof of the Left-Hand Side (LHS)

$$\text{LHS} = \neg[p \rightarrow (\neg q \wedge \neg r)]$$

Implication Law: $p \rightarrow q \equiv \neg p \vee q$

$$\neg[\neg p \vee (\neg q \wedge \neg r)]$$

De Morgan's Law: $\neg(p \vee q) \equiv \neg p \wedge \neg q$

$$\neg\neg p \wedge \neg(\neg q \wedge \neg r)$$

Double Negation: $\neg\neg p \equiv p$

$$p \wedge \neg(\neg q \wedge \neg r)$$

De Morgan's Law: $\neg(p \wedge q) \equiv \neg p \vee \neg q$

$$p \wedge (\neg\neg q \vee \neg\neg r)$$

Double Negation: $\neg\neg p \equiv p$

$$p \wedge (q \vee r)$$

LHS simplified: $p \wedge (q \vee r)$

Proof of the Right-Hand Side (RHS)

$$\text{RHS} = p \wedge [(q \rightarrow r) \rightarrow r]$$

Implication Law: $p \rightarrow q \equiv \neg p \vee q$

$$p \wedge [(\neg q \vee r) \rightarrow r]$$

Implication Law: $p \rightarrow q \equiv \neg p \vee q$

$$p \wedge [\neg (\neg q \vee r) \vee r]$$

De Morgan's Law: $\neg(p \vee q) \equiv \neg p \wedge \neg q$

$$p \wedge [(\neg\neg q \wedge \neg r) \vee r]$$

Double Negation: $\neg\neg p \equiv p$

$$p \wedge [(q \wedge \neg r) \vee r]$$

Distributive Law :

$$(p \wedge q) \vee r \equiv (p \vee r) \wedge (q \vee r)$$

$$p \wedge [(q \vee r) \wedge (\neg r \vee r)]$$

Negation Law $\neg r \vee r \equiv T$

$$p \wedge [(q \vee r) \wedge T]$$

Identity Law: $p \wedge T \equiv p$

$$p \wedge (q \vee r)$$

RHS simplified: $p \wedge (q \vee r)$

Conclusion

$$\text{LHS} = \text{RHS} = p \wedge (q \vee r)$$

The truth table of LHS

$$\text{LHS} = \neg[p \rightarrow (\neg q \wedge \neg r)]$$

p	q	r	$\neg q$	$\neg r$	$(\neg q \wedge \neg r)$	$[p \rightarrow (\neg q \wedge \neg r)]$	$\neg[p \rightarrow (\neg q \wedge \neg r)]$
T	T	T	F	F	F	F	T
T	T	F	F	T	F	F	T
T	F	T	T	F	F	F	T
T	F	F	T	T	T	T	F
F	T	T	F	F	F	T	F
F	T	F	F	T	F	T	F
F	F	T	T	F	F	T	F
F	F	F	T	T	T	T	F

The truth table of RHS

$$\text{RHS} = p \wedge [(q \rightarrow r) \rightarrow r]$$

p	q	r	(q→r)	(q→r)→r	p∧[(q→r)→r]
T	T	T	T	T	T
T	T	F	F	T	T
T	F	T	T	T	T
T	F	F	T	F	F
F	T	T	T	T	F
F	T	F	F	T	F
F	F	T	T	T	F
F	F	F	T	F	F

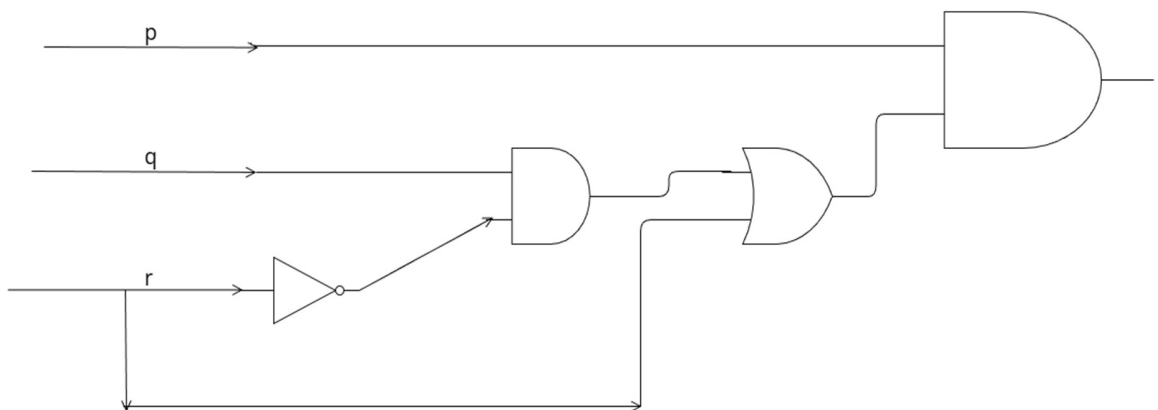
Logic Diagram

$$\text{RHS} = p \wedge [(q \wedge \neg r) \vee r]$$

$$p \wedge (q \vee r) \equiv p \wedge [(q \wedge \neg r) \vee r]$$



$$\text{RHS} = p \wedge [(q \wedge \neg r) \vee r]$$



$$\text{LHS} = p \wedge (q \vee r)$$

$$p \wedge (q \vee r) \equiv p \wedge [(q \wedge \neg r) \vee r]$$



$$\text{LHS} = p \wedge (q \vee r)$$

